

MSDM5004 Spring 2026

Exercise 2

Feb 27

Check the order of convergence of the forward Euler method for solving the ODE

$$y'(t) = e^{t^2}, \quad y(0) = 1$$

at $t = 1$. Choose the largest time step $h = 1/128$.

Solution.

```
n=64;
m=6;
y1=zeros(m,1);
for k=1:m
n=n*2;
dt=1/n;
y=1;
for i=1:n
t=(i-1)*dt;
y=y+exp(t*t)*dt;
end;
y1(k)=y;
end
y1
for k=1:(m-2)
r=(y1(k)-y1(k+1))/(y1(k+1)-y1(k+2))
end
```

Results:

```
y1 =
2.455967359026169
2.459302639641564
2.460975464545025
2.461813173167322
2.462232351522464
2.462442021711122
r =
1.993801388594050
r =
1.996905438161672
r =
1.998453908750579
r =
1.999227252216957
```